

PROJECT DOCUMENTATION WORKSHEET

Introduction to Machine Learning for Cybersecurity Capstone

Team Information

Team #	N/A
Team Member Names	John Flack; team never reached out, completed solo
Capstone GitHub Project Link	[To be added after repository creation]

Project Summary - CRISP-DM Framework

1. Business Understanding

Problem statement	Phishing websites are a cybersecurity threat because they imitate legitimate sites and attempt to steal credentials, financial information, or other sensitive data. This project asks whether machine learning can classify websites as phishing or legitimate based on structured website features.
Project goals and success criteria	The project goal is to build a supervised machine learning workflow that loads a phishing dataset, prepares target and features, trains multiple models, evaluates performance, visualizes results, and explains the cybersecurity meaning of model errors. Success is measured using accuracy, precision, recall, F1 score, confusion matrix results, and ROC-AUC.

2. Data Understanding

Dataset used	UCI Machine Learning Repository - Phishing Websites Dataset. The dataset contains 11,055 website records and 30 integer-coded features. The target identifies whether a website is phishing or legitimate.
Key variables and initial observations	The features describe website characteristics such as URL behavior, domain indicators, SSL state, redirects, links, and other phishing-related signals. The original target variable is Result, which is mapped into a binary is_phishing target for modeling.

3. Data Preparation

Main cleaning and transformation steps	The ARFF dataset is parsed into a pandas DataFrame. Result is converted so that phishing = 1 and legitimate = 0. Feature columns are separated from the target, missing values are checked, and a stratified train/test split is created. Logistic Regression uses StandardScaler through a pipeline; Random Forest is trained on the original integer-coded features.
Final dataset used for modeling	The final modeling dataset includes the 30 website feature columns as X and the binary is_phishing target as y.

4. Modeling

Algorithms applied	Logistic Regression and Random Forest Classifier.
Reason for model selection	Logistic Regression provides an interpretable baseline model. Random Forest provides a stronger nonlinear ensemble model and supports feature importance analysis. Using both models allows comparison between a simpler baseline and a more flexible classifier.

5. Evaluation

Best model and	Best model: Random Forest Classifier.
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performance metrics	Random Forest performance on the test set: accuracy = 97.06%, precision = 96.64%, recall = 96.73%, F1 score = 96.69%, and ROC-AUC = 0.9964. For comparison, Logistic Regression achieved accuracy = 93.17%, precision = 92.95%, recall = 91.53%, F1 score = 92.24%, and ROC-AUC = 0.9786.
Does the model meet project goals? Why or why not?	Yes. The project goal was to build and evaluate a supervised machine learning workflow for phishing detection. The Random Forest model met this goal because it produced strong precision, recall, F1 score, and ROC-AUC while keeping both false positives and false negatives relatively low. The model should still be treated as analyst decision support rather than a fully autonomous blocking system: final liability rests with the human.

6. Insights and Next Steps

Key analytical insight	The Random Forest model performed best and identified important phishing indicators such as SSLfinal_State, URL_of_Anchor, web_traffic, having_Sub_Domain, and Prefix_Suffix. The key security insight is that false negatives are the most serious error type because they represent phishing websites that the model incorrectly treats as legitimate. In the best model, the confusion matrix showed 1,198 true negatives, 33 false positives, 32 false negatives, and 948 true positives.
Next steps	Future work could include hyperparameter tuning, cross-validation, testing on newer phishing datasets, threshold tuning, model monitoring, and explainability methods such as SHAP or LIME so analysts can better understand why a website was classified as phishing. That's outside the scope of this class.

Methods Overview

Key data cleaning steps	Parsed ARFF data, checked missing values, mapped target labels, separated features and target, and created a stratified train/test split.
Feature engineering strategy	The initial project uses the provided structured website features without creating new variables. The primary transformation is target remapping. Scaling is applied only where needed for Logistic Regression.
Models applied	Logistic Regression and Random Forest Classifier.

Results

Best model and performance metrics	Best model: Random Forest Classifier. Accuracy = 97.06%, precision = 96.64%, recall = 96.73%, F1 score = 96.69%, and ROC-AUC = 0.9964. Confusion matrix: TN = 1,198, FP = 33, FN = 32, TP = 948.
Key predictive features	The most important Random Forest features were SSLfinal_State, URL_of_Anchor, web_traffic, having_Sub_Domain, and Prefix_Suffix. Logistic Regression also highlighted URL_of_Anchor, Prefix_Suffix, SSLfinal_State, SFH, and Links_in_tags as highly influential.
One major analytical insight	The model can support phishing detection by prioritizing suspicious websites for analyst review, but high performance metrics do not eliminate risk. Missed phishing websites remain a serious concern because a false negative could expose users to credential theft, malware (C2, etc.), or fraud.

Reflection

Biggest technical challenge encountered	The biggest technical challenge was selecting a dataset that was cybersecurity-relevant, manageable for a course capstone, and distinct from the previous credit card fraud labs dataset (which would have been an easy . . .but known . . .quantity). Another challenge was interpreting results in a security context rather than only reporting high accuracy.
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How the challenge was solved	The challenge was solved by using the UCI Phishing Websites Dataset, which is structured, directly related to cybersecurity, and appropriate for binary classification. The project also compared Logistic Regression with Random Forest so the final analysis included both an interpretable baseline and a stronger nonlinear model.
One improvement for future work	Future work should test the model against newer phishing URLs, tune decision thresholds, use cross-validation, and add explainability tools so analysts can better understand why a website was classified as phishing.

Team Contribution

Role of each team member	John Flack: dataset selection, project design, notebook development, exploratory data analysis, data preparation, model training, evaluation, visualization, interpretation, and documentation. No team members reached out, or attempted to, thus, the project was completed solo in order to maintain work standards and my own grade in the class.
Confirmation of collaborative work	N/A, completed solo.